



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SAMUNDI FUEL AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

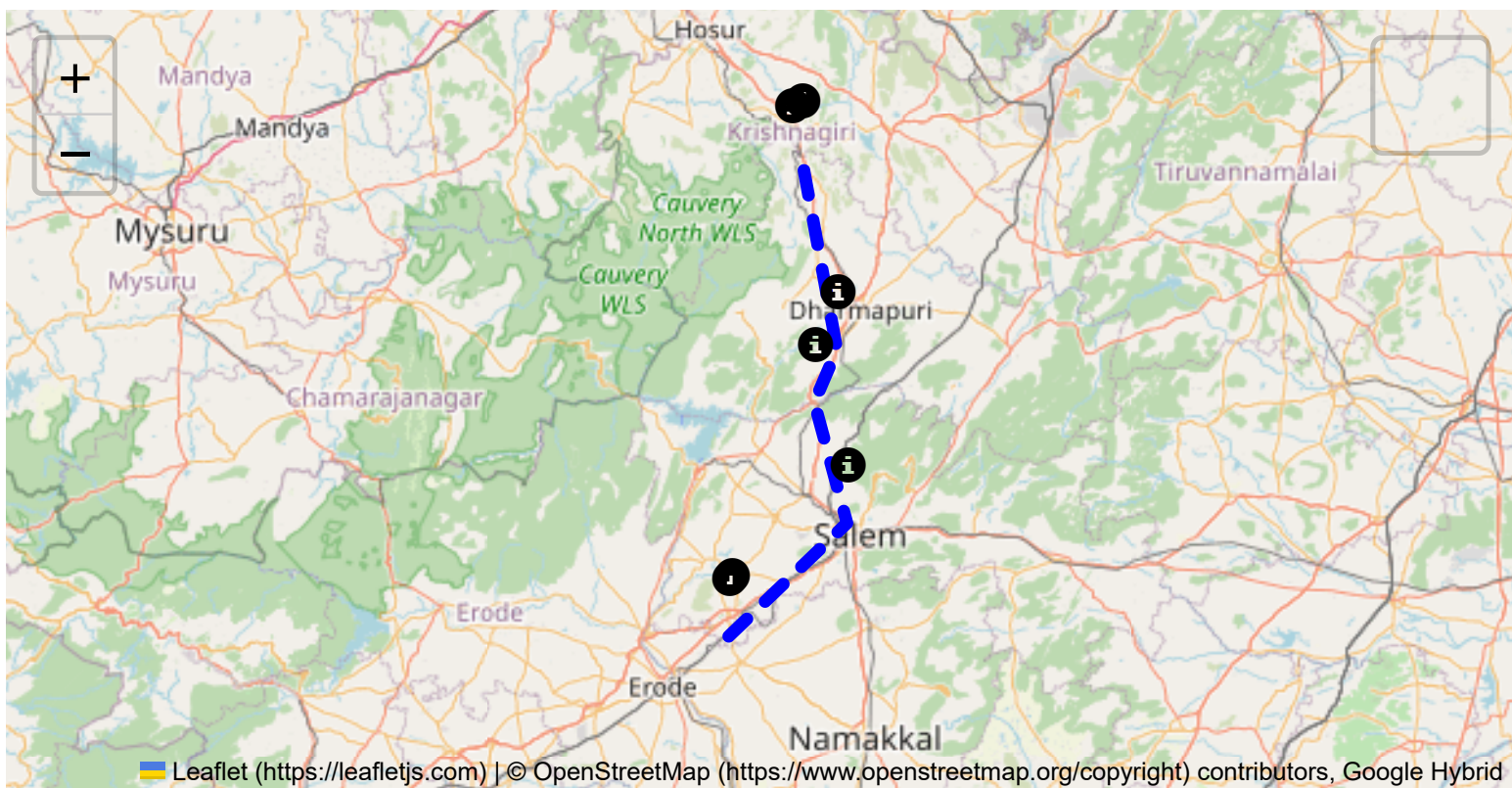
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

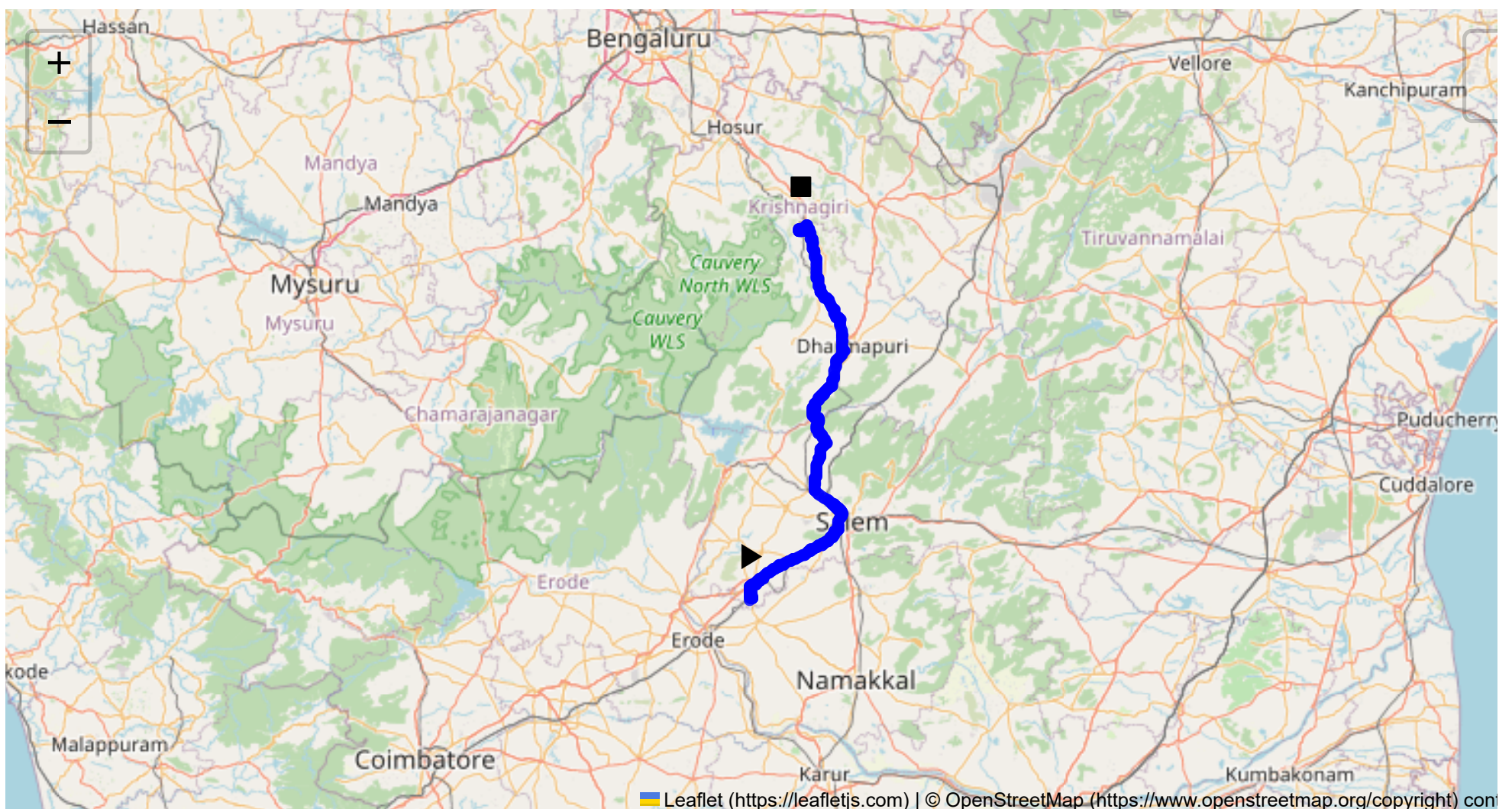
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 147.85 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.448076, 78.006449)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route begins in Sangagiri, Tamil Nadu, and proceeds northeast towards Ekkandahalli. Key waypoints include Kangeyampalayam, Salem, Thoppur, Seshampatti, and Kaduchettipatti, covering major roads such as SH 17 and the Bangalore Bypass Road. The journey involves traversing both rural and semi-urban areas, highlighting diverse driving conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. This route may encounter:

- **Monsoon Season (June to September):** Heavy rains can lead to waterlogging and reduced visibility.
- **Summer (March to May):** High temperatures may affect vehicle performance.
- **Cyclone Season (October to December):** Occasional cyclones can disrupt travel and cause road damage.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8-10 AM) and Evening (5-7 PM), especially near Salem and Thoppur.
- **Congestion-Prone Areas:** Salem city center and junctions near major highways.

4. Assessment of Road Quality and Infrastructure

- **Sangagiri to Kangeyampalayam:** Good road conditions with minimal potholes.
- **Salem Region:** Well-maintained infrastructure, though congested.
- **SH 17 towards Kaduchettipatti:** Varies from good to moderate, with occasional rough patches requiring cautious driving.

5. Suggestions for Alternative Routes for Emergencies

- **From Salem:** Consider NH 44 for bypassing congested areas.
- **For Kaduchettipatti:** Use local roads for detours if SH 17 is blocked.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permission Required:** Ensure proper permits for carrying hazardous materials.
- **Restricted Times:** Movement may be restricted during peak hours in urban areas.
- **Safety Protocols:** Strict adherence to safety and security protocols mandated by local authorities.

7. Overview of Historical Incidents

While no major incidents are associated specifically with these points, southern Tamil Nadu can be susceptible to:

- **Accidents in Rural Areas:** Due to less enforced traffic rules.
- **Hazmat Incidents:** Ensure thorough checks given incidents involve poor handling procedures.

8. Environmental Considerations and Sensitive Areas

- **Protected Zones:** Closely monitor sections near wildlife reserves or protected agricultural zones.

- **Noise Restrictions:** Be respectful of noise levels near residential areas and schools.

9. Analysis of Communication Coverage

- **Good Coverage:** Near major towns such as Salem.
- **Potential Dead Zones:** Rural stretches particularly around agricultural areas between Kangeyampalayam and Thoppur.

10. Estimated Emergency Response Times

- **Salem:** Rapid; within 15-20 minutes due to urban infrastructure.
- **Rural Areas:** Could range from 30 minutes to over an hour depending on specific location and conditions.

12. Overall Summary of Risk Assessment

This route offers various challenges, from weather to traffic congestion. While infrastructure remains generally supportive of heavy vehicles, permit adherence and strategic planning are crucial. Communication gaps and emergency response time variability should be factored into logistical considerations. For safety, continuous monitoring of conditions and potential adjustments are advised, alongside compliance with local regulations governing hazardous materials.

Consider employing advanced GPS systems for real-time updates and establishing communication protocols with dispatchers, particularly across identified communication dead zones. Overall, with proper preparation, the route presents a moderate risk level.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44028, 77.87541	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45007, 77.87201	15 KM/Hr
4	Turn	Medium	11.46392, 77.87328	30 KM/Hr
5	Turn	High	11.47391, 77.86844	15 KM/Hr
6	Turn	High	11.49148, 77.89601	15 KM/Hr
7	Turn	High	11.49206, 77.89601	15 KM/Hr
8	Turn	Medium	11.53501, 77.95707	30 KM/Hr
9	Turn	Medium	11.53601, 77.95856	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.62403, 78.10893	30 KM/Hr
11	Turn	High	11.64180, 78.11808	15 KM/Hr
12	Turn	High	11.64325, 78.11993	15 KM/Hr
13	Turn	Medium	11.67943, 78.12778	30 KM/Hr
14	Turn	Medium	11.69508, 78.10787	30 KM/Hr
15	Turn	Medium	11.74118, 78.05037	30 KM/Hr
16	Turn	Medium	11.75776, 78.05015	30 KM/Hr
17	Turn	Medium	11.78039, 78.05792	30 KM/Hr
18	Turn	Medium	11.86753, 78.08331	30 KM/Hr
19	Turn	Medium	11.86981, 78.08387	30 KM/Hr
20	Turn	Medium	11.88041, 78.08143	30 KM/Hr
21	Turn	Medium	11.91302, 78.05856	30 KM/Hr
22	Turn	Medium	11.93245, 78.06456	30 KM/Hr
23	Turn	Medium	11.95594, 78.04730	30 KM/Hr
24	Turn	Medium	11.95818, 78.04978	30 KM/Hr
25	Turn	Medium	11.95876, 78.04990	30 KM/Hr
26	Turn	High	11.96035, 78.04946	15 KM/Hr
27	Turn	Medium	11.96304, 78.05136	30 KM/Hr
28	Turn	Medium	11.96399, 78.05135	30 KM/Hr
29	Turn	Medium	11.96458, 78.05084	30 KM/Hr
30	Turn	Medium	11.96496, 78.04988	30 KM/Hr
31	Turn	High	11.96580, 78.04923	15 KM/Hr
32	Turn	Medium	11.96675, 78.04939	30 KM/Hr
33	Turn	Medium	11.96791, 78.05070	30 KM/Hr
34	Turn	Medium	12.12050, 78.13284	30 KM/Hr
35	Blind Spot	Blind Spot	12.46203, 78.02991	10 KM/Hr
36	Turn	High	12.45346, 78.02511	15 KM/Hr
37	Turn	Medium	12.45410, 78.02305	30 KM/Hr
38	Turn	Medium	12.45302, 78.02087	30 KM/Hr
39	Turn	Medium	12.44969, 78.01876	30 KM/Hr
40	Turn	High	12.45085, 78.00733	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
7	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
8	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
11	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
12	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
13	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
16	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
20	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
21	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
22	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
23	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
25	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
26	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
27	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
28	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
29	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
31	pharmacy	Salem Medicals	11.7343192, 78.0566844	30 km/h	Medium
33	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
34	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
36	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
38	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
39	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
40	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
42	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
17	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
18	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
19	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
24	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
30	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
32	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
37	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
41	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45007, 77.87201



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46392, 77.87328



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47391, 77.86844



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49148, 77.89601



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49206, 77.89601



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64325, 78.11993



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67943, 78.12778



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69508, 78.10787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74118, 78.05037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75776, 78.05015



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78039, 78.05792



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86753, 78.08331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86981, 78.08387



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88041, 78.08143



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91302, 78.05856



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93245, 78.06456



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95594, 78.04730



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95876, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96035, 78.04946



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96304, 78.05136



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96458, 78.05084



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96496, 78.04988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96675, 78.04939



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12050, 78.13284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.46203, 78.02991



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.45346, 78.02511



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.45410, 78.02305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.45302, 78.02087



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.44969, 78.01876



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.45085, 78.00733
